

Fingerprint Datasets — Acquired and Unavailable

Enhancement of fingerprints degraded by occupational wear, ageing and skin damage Master's thesis · Astitva Srivastava · 11 September 2026

A record of which datasets have been obtained and which have not. Image counts and sizes are measured from per-image manifests ([docs/datasets/manifests/](#)), not quoted from the publishers. Live access status is tracked in [docs/datasets/REGISTRY.md](#) .

Summary: 10 datasets acquired, 38,973 images, 2.2 GB. One further set (SOCOFing) was downloaded and has been excluded — see §3. Eight datasets are pending; five are permanently unavailable.

1. Acquired — in use

Dataset	Images	MB	Image size	dpi	Structure
Anguli corpus (generated here)	22,336	654	275×400	500	5,584 fingers × 4
L3-SF	8,140	569	320×240 / 512×512	1200 (partial)	740 identities
DTD textures	5,640	633	variable	n/a	backgrounds, not prints
MINEX validation	801	114	167×325 – 406×662	500	54 subjects
Neurotechnology U.are.U	520	62	326×357	512	65 fingers × 8
Neurotechnology CrossMatch	408	39	504×480	500	51 fingers × 8
FVC2004 "B" (4 databases)	320	55	288×384 – 640×480	500 / 512	40 fingers × 8
FVC2002 "B" (4 databases)	320	41	300×300 – 296×560	500 / 569	40 fingers × 8
FVC2000 "B" (4 databases)	320	38	240×320 – 448×478	500	40 fingers × 8
FVS	168	33	256×256	unstated	21 fingers × 8

FVC2000 / FVC2002 / FVC2004 — "B" subsets

<http://bias.csr.unibo.it/fvc2000/download.asp> (also [/fvc2002/](#) , [/fvc2004/](#)) University of Bologna. Free for research; the larger "A" subsets ship only with the *Handbook of Fingerprint Recognition*.

Three biennial fingerprint verification competitions, four sensor databases each — optical, capacitive, thermal sweep, and synthetic. All twelve "B" databases held: 960 images, 10 fingers × 8 impressions each. FVC2002 and FVC2004 deliberately instructed subjects to give *difficult* impressions, including wet and dry fingers, which makes **FVC2004 DB1 the closest freely available analogue to occupationally worn prints**. This is the most widely reported benchmark in the enhancement literature and the primary real-data evaluation set here.

Neurotechnology sample databases

<https://www.neurotechnology.com/download.html#databases> Free vendor evaluation data.

Two clean, consistent optical captures: CrossMatch Verifier 300 (51 fingers $\times$ 8) and U.are.U 4000 (65 fingers $\times$ 8). Eight impressions per finger gives 1,428 and 1,820 genuine comparison pairs respectively — the largest genuine-pair counts in the corpus, and therefore the tightest error-rate estimates. Used as the clean control condition: a method that damages good images fails regardless of how it performs on poor ones.

MINEX validation imagery

<https://github.com/usnistgov/minex> NIST, public domain.

801 raw images from NIST's MINEX III template-conformance programme, 54 subjects. The images ship without metadata, but the validation driver's source header carries width, height, **quality band and finger position** for all 801, which we parse. That gives a finger-position and quality stratification axis — 8 positions, 132 images labelled poor or fair — on permissively licensed data.

Anguli-generated paired corpus (generated locally)

<https://dsl.cds.iisc.ac.in/projects/Anguli/> (generator) Ansari, *Generation of Synthetic Fingerprint Databases*, MSc thesis, IISc Bangalore, 2011. Generator free for research; the corpus is our own output and carries no restriction.

5,584 synthetic fingers, each with one clean master plus three simulated impressions — 22,336 images at 275×400 . **The training set.** Supervised enhancement needs clean/degraded pairs, and this is the only source of perfectly paired data available; the clean master is the reconstruction target and our own degradation code produces the input. Requested at 20,000 fingers, interrupted at 5,584 by a memory exhaustion; the directory is named `anguli_dev_5584` so the size is not mistaken. 5,584 is ample — the 2026 state of the art was trained on 360 images.

DTD — Describable Textures Dataset

<https://www.robots.ox.ac.uk/~vgg/data/dtd/> Cimpoi et al., *Describing Textures in the Wild*, CVPR 2014. Free for research.

5,640 texture photographs in 47 categories. Not fingerprint data — this is the surface-background bank used to build the *control* degradation model (conventional latent-style degradation), against which our wear model is compared. Partitioned by category so no test-set texture can reach training, which answers Cappelli's leakage criticism of DNUNets and FingerGAN.

L3-SF — Level-3 Synthetic Fingerprints

<https://github.com/luannd/L3-SF> Wyzykowski, Segundo & Lemes, ICPR 2020. CC BY-NC-SA 4.0.

8,140 synthetic fingerprints with level-3 detail — sweat pores and scratches — including a 740-image subset with pore coordinate annotations. Secondary: level-3 detail sits above the 500 dpi civil-capture regime this thesis targets. Note the advertised 1200 dpi applies only to the 740 annotated 512×512 images; the other 7,400 are 320×240 and cannot be 1200 dpi.

FVS

<http://fvs.sourceforge.net/> — open-source project sample data. 168 images, 21 fingers $\times$ 8, 256×256 , resolution unstated. Minor supplementary set.

2. Acquired but excluded

SOCOFing

<https://www.kaggle.com/datasets/ruizgara/socofing> — Shehu et al., arXiv:1807.10609, 2018.

55,270 images (6,000 real prints from 600 subjects, plus 49,270 synthetically altered versions with damage-type and severity labels). Downloaded, but **excluded from the project on two grounds**. First, it is Kaggle-hosted, and supervisor guidance is that Kaggle datasets are not to be used for training. Second, and independently, our own measurement does not support its quality claim: the images are 96×103 px and declared 500 dpi, which would describe a finger 4.9 mm across — the true sampling is nearer 200 dpi, far below civil capture resolution. Its damage is also synthetic, so it does not represent the target population regardless.

The severity axis it would have provided is instead obtained by applying our own wear model at controlled severities to real clean prints, which gives known rather than undocumented severity labels. The loss is demographic stratification, for which no substitute is currently held.

3. Not acquired

3.1 Registration or approval pending

Dataset	Link	What it is	Why not yet
CASIA-FingerprintV5	http://biometrics.idealtest.org (id 7)	500 subjects $\times$ 8 fingers $\times$ 5 impressions. The population explicitly includes workers and waiters , and difficult impressions were requested — the closest free analogue to an occupationally worn population.	Registration plus manual human approval. Confirmed live 10 Sep 2026; walkthrough in <code>docs/datasets/casia-access.md</code> . Now the highest-priority outstanding item.
CASIA Fingerprint Subject Ageing	http://biometrics.idealtest.org (id 15)	3,664 images. Public longitudinal fingerprint data is close to non-existent, and ageing is a named degradation cause in this thesis.	Same registration route. Found via the portal API; it appears in neither the thesis plan nor the literature review.
PrintsGAN	MSU — Engelsma, Grosz & Jain	525,000 synthetic images, 35,000 fingers, multiple impressions per identity. Large-scale pretraining.	Signed agreement sent to Steven Grosz, MSU; awaiting reply.
Tsinghua SD27 annotations	http://ivg.au.tsinghua.edu.cn/dataset/NIST.php	Manually labelled orientation fields, period maps, skeletons and ROIs — the class of supervision Cappelli's orientation and frequency networks were trained on.	Request not yet sent. Annotates SD27 images, which are themselves withdrawn (§3.3).
FVC manual segmentation ground truth	Thai, Huckemann & Gottschlich, <i>PLOS ONE</i> , 2016	Hand-marked segmentation masks for FVC images — what Cappelli trained his segmentation on.	Request not yet sent.

3.2 Licensed — institutional signature required

All route through IIIT-Delhi's Image Analysis and Biometrics lab (databases@iab-rubric.org) or NIST, and all need a signature above student level.

Dataset	What it is	Why not yet
Rural Indian Fingerprint DB	Rural and urban Indian subjects with genuine occupational degradation — the target population of this thesis , and the benchmark of the Joshi research line.	Requires a Registrar / Head-of-Institution signature . Request drafted in docs/correspondence/ , not yet submitted. Longest lead time of anything here.
MUST	~21,000 impressions with manually marked minutiae , pore counts and semantic masks. The dataset that would allow true minutiae F1 against expert ground truth.	Same signature route; drafted, not submitted.
IIIT-D MOLF	19,200 images, 100 Indian subjects, five capture methods. Used by the Joshi line and MLFGNet.	Same signature route; drafted, not submitted.
NIST SD302 (N2N) and SD300	SD302: multi-sensor live capture, 200 subjects × 10 fingers. SD300: 888 subjects of naturally poor, uncooperative ink-card captures.	Institutional email and manual NIST review. Not submitted.

Because these are slow and uncertain, the thesis is structured so that no phase depends on any of them. The Rural Indian DB is treated as a *confirmatory* external test set: valuable if it arrives, not load-bearing if it does not.

3.3 Permanently unavailable

Dataset	Status	What this means
NIST SD27	Withdrawn January 2017	258 real crime-scene latents with expert-validated minutiae. This is the benchmark of essentially the entire latent-enhancement literature, <i>including the 2026 state of the art</i> , which still reports on it because established groups hold legacy copies. Its published numbers cannot be reproduced by anyone without one — a documented problem in the field, and the reason this thesis builds its protocol on obtainable data instead.
ChaLearn LAP Track 3	Provider broken; unrecoverable	84,000 paired clean/degraded images, the intended primary training set. Sign-up returns HTTP 500 and password reset fails; several email addresses tried. Replaced by generating clean masters with Anguli and reimplementing ChaLearn's nine degradation artefacts in our own code (ADR 0003).
NIST SD4 / SD9 / SD10	Discontinued	Historical baselines cannot be reproduced.
NIST SD14	Uncertain; treated as unavailable	Was the 27,000–30,000-image training source for DNUNets, FingerGAN and SFP, so those training sets cannot be reconstructed either.

4. Licence note

IAB, NIST, CASIA and FVC-A data may not be redistributed. Images are never committed to the repository — only manifests recording each file's SHA-256, dimensions and declared resolution. Of the acquired holdings, only the Anguli corpus (our own output) and MINEX (public domain) could be released alongside the thesis.

Figures measured from committed manifests via `scripts/dataset_dossier_stats.py`. Access status per `docs/datasets/REGISTRY.md`.